ECUE 3.1.3: Neural Communication

From Biological Mechanisms to Artificial Computational Architectures

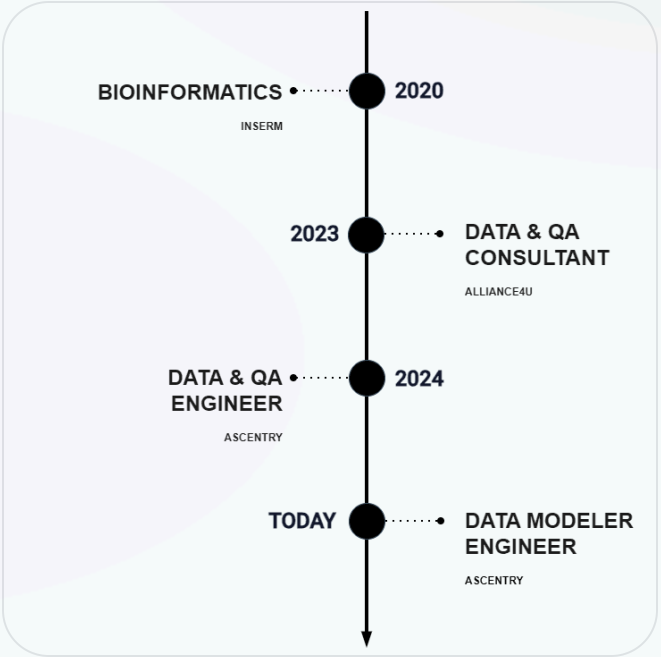
CESI Bachelor AI (Year 2)

Jérémy MARTIN • MSc in Bioinformatics and Systems Biology

Introduction

Who Am I?

- **Background and Research**
  - **MSC IN BIOINFORMATICS & SYSTEMS BIOLOGY (2020)**
  - **3 YEARS AT INSERM**: Data pipelines, Transcriptomics analytics & ML on multiplexed fluorescent imaging.
- **Data Engineering & Medtech**
  - **DATA ENGINEER & MODELER AT ASCENTRY**: Data architecture, regulatory-compliant processes, ML & NLP on customer & marketing data



Ice Breaker: "What Is a Neuron?"

Dynamic Exchanges

- When you hear the word **neuron**, what comes to mind?
- Do you think a neuron is more like:
  - A simple electrical switch (ON/OFF)?
  - A tiny analog computer performing calculus?
  - A bio-chemical factory?
- How many neurons and synapses are there in the human brain?

Key Brain Figures

The brain is **AWESOME**

- $\pm 86 \times 10^9$ neurons *(86 billion)*[*]
- $\pm 10^{14}$ synapses[*]
- Average connectivity: $\approx 7,000$ synaptic inputs per neuron
- Power consumption: ≈ 20 Watts

SOURCE[*]: HTTPS://PMC.NCBI.NLM.NIH.GOV/ARTICLES/PMC11423976/

Course Roadmap

Course will be split into two parts:

Part 1: Foundations

- 🧭 **1.** Introduction & Epistemological Bridge
  - *Scope and core concepts*
- 🦾 **2.** The Biological Neuron
  - *Conceptual challenge: What is a neuron?*
- f_x **3.** From Biology to Mathematics
  - *Abstraction and Information encoding*
- </> **4.** Logical Gates & McCulloch-Pitts Neurons
  - *Live coding: simulation on NumPy*

Part 2: Architectures & Learning

- 🧠 **5.** The First Learning Neuron
  - *From static M&P model to plasticity*
- 📈 **6.** Limits & Evolution
  - *The XOR Crisis*
- 🖧 **7.** Multi Layer Perceptrons
  - *Non-linearity, feedforward and communication*
- 🚩 **8.** Synthesis & Closure
  - *Closing the Epistemological Bridge*

The Week at a Glance

	Morning	Afternoon
Monday	Ice breaker, the module, the BRIDGE, the projects *(3h)*	**Ch. 2** The Biological Neuron, start of **Ch. 3**, up to the Python warm-up *(3h)*
Tuesday	**Ch. 3** From Biology to Mathematics *(3h30)*	**Ch. 4** Logic gates workshop, start of **Ch. 5**: plasticity *(3h30)*
Wednesday	Projects: autonomous work, questions *(2h)*	**Ch. 5** The First Learning Neuron, start of **Ch. 6** *(3h30)*
Thursday	**Ch. 6** The XOR Crisis, start of **Ch. 7** *(3h30)*	**Ch. 7** and **Ch. 8** Closure *(1h)*, then Projects: autonomous work *(2h)*

> **21h of course + 4h of project work**, in autonomy, with me around for your questions. **Project article + notebook: Friday 20 November. Presentation: Monday 23 November.** Every chapter mixes lecture, **paper & pencil** exercises and **live coding** in Jupyter.

Materials: Everything Lives on GitHub

Repository: [martin-jeremy/cesi-2026-neuronal_communication]

What you will find there *(available now, or added as the course goes)*

- `slides/`: the slides of the course
- `notebooks/`: almost one notebook per chapter
  - `ChapterN_LiveExercises.ipynb`: the exercises
  - `ChapterN_LiveExercises_SOLUTIONS.ipynb`: the solutions
- `projects/`: project subjects, datasets and helpers

Get ready, once

```
git clone <repository-url>
cd <repository>
uv sync                 # installs the environment
uv run jupyter lab      # opens Jupyter
```

Then run `notebooks/Chapter1_Setup.ipynb`.

Solutions are there to **check** your work, not to replace it: always try first, on paper, then in code.

Part 1: Foundations • Chapter 1

Introduction & Epistemological Bridge

What This Module IS and IS NOT

Before starting this module, it's important to understand what this course is and is not.

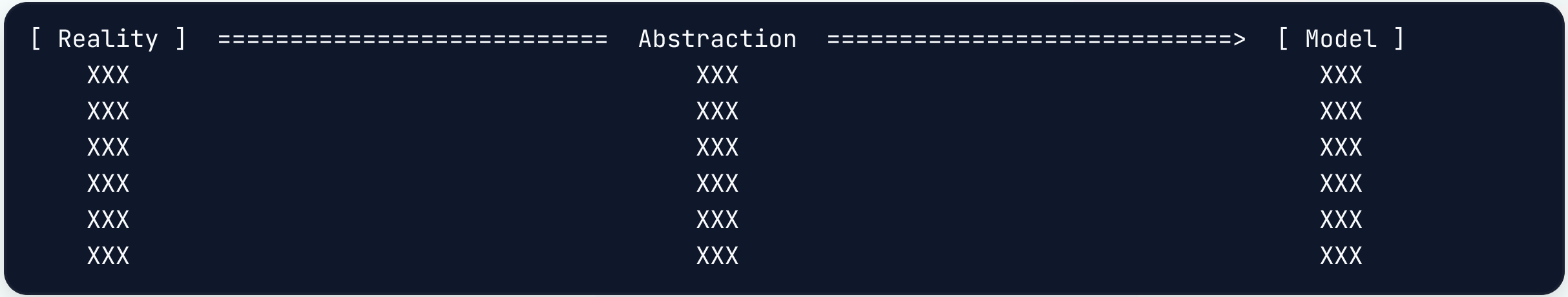
What It Is NOT

- **Not a pure Neuroscience course:** We do not memorize hundreds of anatomical regions, molecular pathways, or clinical pathologies.
- **Not a black-box Deep Learning course:** We will not simply call `import tensorflow` or `torch.nn` to train massive models without inspecting internal mechanics.

What It IS

- **The Formal Bridge:** Connecting biological electrical/chemical communication principles to formal mathematical abstractions.
- **First-Principles AI:** Understanding *why* weights, thresholds, activations, and architectures are designed the way they are.
- **Scratch Implementation:** Building the core computational blocks in pure Python/NumPy.

The Epistemological Bridge



> **Definition:** The **EPISTEMOLOGICAL BRIDGE** is a conceptual framework that maps validated knowledge from one domain onto another.

The Epistemological Bridge

The Map Example

A map is a good way to picture what an epistemological bridge does.

This is a satellite view:



This is a **REALITY**

Each element can be **any** representation:

- coordinates
- color
- density

This is an **ABSTRACTION**

This is a roadmap:



This is a **MODEL**

The Epistemological Bridge

The Map Example

The bridge is a map, not a mirror!
An epistemological bridge is not a perfect **lift-and-shift copy**.

It's an **abstractive** representation of the reality translated into a new system of knowledge, one that remains **coherent** and **consistent** within its own **mathematical** and **logical** space.

The Epistemological Bridge

In our case, the epistemological bridge that will be our **RED LINE** during all the course is the mapping between biological reality and computational model.

```
[ Biological Reality ]  ======================  Abstraction  ======================>  [ Computational Model ]
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
        XXX                                      XXX                                      XXX
```

> **Objectives:** At the end of this class, you will be able to fill in the missing parts of the mapping and understand where biological inspiration yields powerful algorithms and where artificial engineering diverges for computational efficiency.

Project Briefs: Two Case Studies

You will be split into **two groups**. Both projects are **case studies** of the Epistemological Bridge, through hands-on implementation and analysis.

Project 1: Neural Decoding for a BCI*

- Spike train dataset: biological neurons answering two stimulus classes
- Implement an artificial neuron from scratch (NumPy)
- Train, evaluate, visualize convergence and learned weights
- **Critical analysis:** what the model captures vs. what it discards from the biological signal

BONUS: *what would need to change in the pipeline to account for what was discarded?*

Project 2: MET* & Robotic Prosthesis

- Spike train dataset: biological neurons answering a motor effort test
- Start with a pre-trained model that worked on a previous experiment
- Identify why it doesn't work in this new case, and repair it
- **Critical analysis:** what does each function change biologically and computationally

BONUS: *does it matter where you put activations if you mix them across layers?*

BCI: *Brain Computer Interface*

MET: *Motor Effort Test*

Projects: What You Need, and When You Get It

To succeed in the projects, you need to…	You get it in	When
Read spike trains: rasters, PSTH, firing rates	Ch. 2 and 3	Mon PM, Tue AM
Turn a trial into a vector of numbers (features)	Ch. 3	Mon PM, Tue AM
Code a neuron, then a perceptron, in NumPy	Ch. 3 to 5	Tue, Wed PM
Evaluate a decoder: train / test split, accuracy, standardization	Ch. 5	Wed PM
Handle more than two classes, layers and activations	Ch. 7	Thu
Criticize your model: what is kept, what is discarded	THE BRIDGE	every chapter

Wednesday morning and **Thursday afternoon**: project time, in autonomy, with me around for your questions. **Article + notebook: 20 November. Presentation: 23 November.**

Evaluation modalities: a mini article (4 pages max), a jupyter notebook and a 30-min presentation followed with questions.
Articles and notebooks due-date: Friday 20 Nov. **Presentation:** Monday 23 Nov.

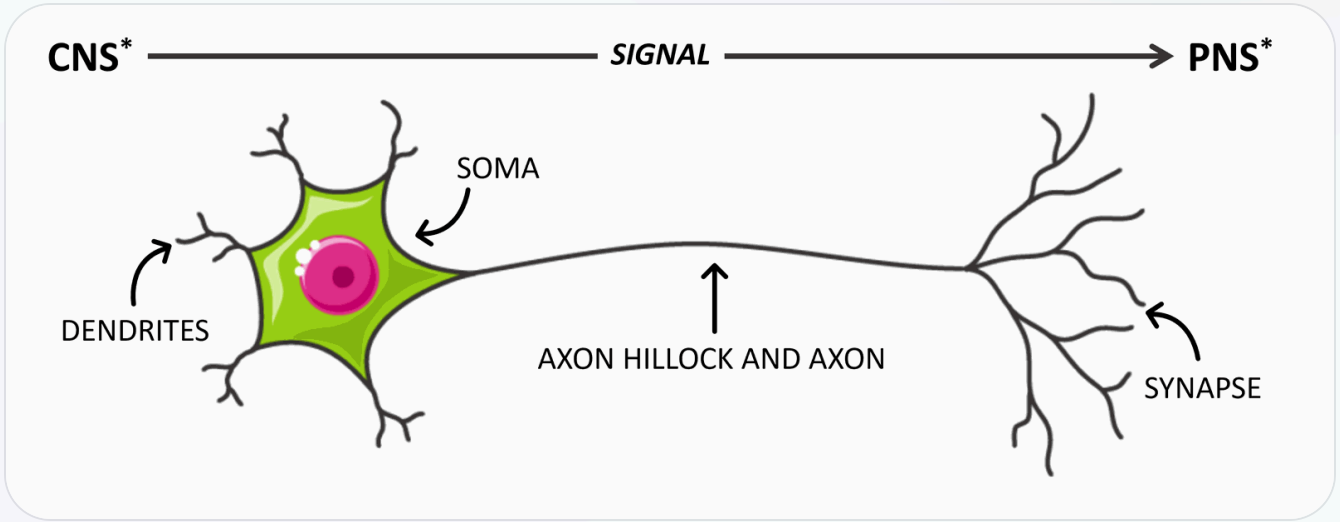
Part 1: Foundations • Chapter 2

The Biological Neuron

Neuron Structure: Overview of a Polarized Anatomy

*Biological neurons are **electrically** excitable cells specialized for receiving, integrating, and transmitting information*

Neuron signaling is **strictly directional** *(polarized)*: information is received by the **dendrites** *(input pole)*, passes through the **soma**, and evaluated at the **axon hillock** before being transmitted along the **axon** to the **synaptic terminals** *(output pole)*.



Some classes of neurons can also transmit information in the opposite direction *(sensory neurons, nociceptors...)*, **but within a single neuron, the signal propagates in one direction only.**

***CNS**: Central Nervous System*

***PNS**: Peripheral Nervous System*

Neuron Structure: Soma & Dendrites

Dendritic Arborization

- Highly branched tree-like extensions often covered with thousands of **dendritic spines**.
- Act as the **input antennas** of the cell.
- Receive chemical neurotransmitters from presynaptic neurons and generate localized postsynaptic potentials (PSPs).

The Soma (Cell Body)

- Metabolic and genetic core containing the nucleus and organelles.
- Acts as a continuous **spatial and temporal integrator**: sums hundreds of incoming graded electrical potentials flowing from dendrites to the axon hillock.

Neuron Structure: Axon & Terminals

Axon & Hillock

- **Axon Hillock:** High density of voltage-gated Na^+ channels; acts as **the decision threshold trigger zone**.
- **Axon:** Long cylindrical projection (up to 1 meter in humans) conducting impulses away from soma.

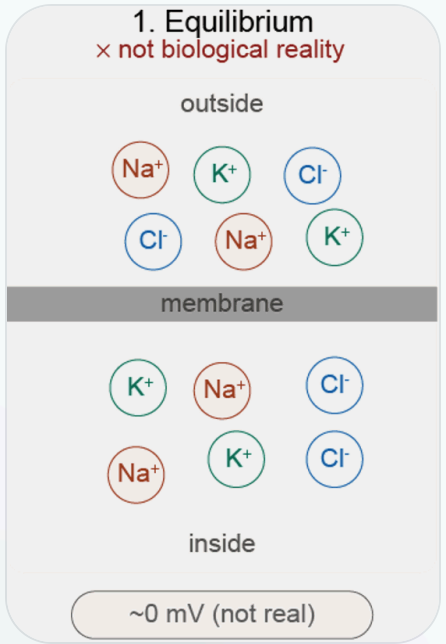
Synaptic Terminals

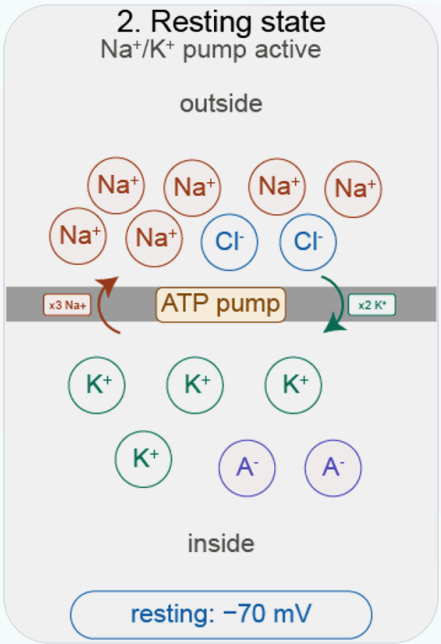
- Specialized terminal boutons containing synaptic vesicles filled with neurotransmitters.
- **Converts electrical pulses into chemical signals** across the synaptic cleft.

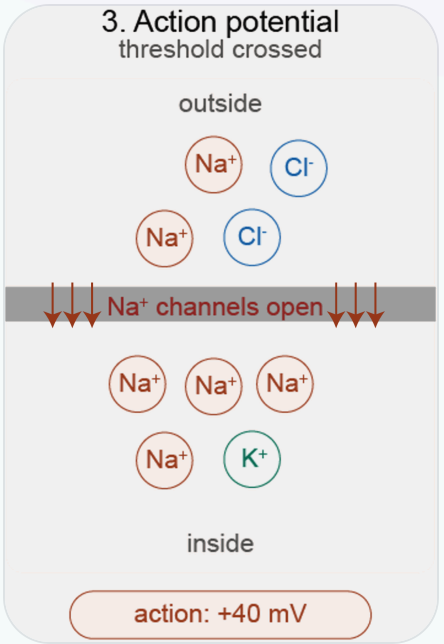
Resting Membrane Potential

But how can a neuron "create" an electrical signal?

→ Thanks to a **gold-standard** engineering pump present in almost ALL our cells!







Resting Membrane Potential

But how can a neuron "create" an electrical signal?

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The Synapse: Where Neurons Communicate

A chemical synapse, step by step

1. The spike reaches the **terminal**
2. **Vesicles** release neurotransmitters into the synaptic cleft
3. They bind **receptors** on the next neuron's dendrite
4. Ion channels open: a small, local potential appears, the **PSP**

Excitatory synapse *(e.g. glutamate)*: an **EPSP**, the membrane moves **toward** the threshold.

Inhibitory synapse *(e.g. GABA)*: an **IPSP**, the membrane moves **away** from the threshold.

Not all synapses are equal: the amount of neurotransmitter and the number of receptors set the **SYNAPTIC STRENGTH**, and this strength can **change** over time.

In Summary:

Neural Communication is nothing more than a Signal Pipeline

1. **Intent**: *An upstream signal reaches the neuron (from the brain or sensory organs).*
2. **Reception**: *Dendrites receive signals and convert them into local electrical shifts.*
3. **Integration & Propagation**: *The soma sums these signals until the threshold is reached at the axon hillock, then an electrical impulse travels down the axon.*
4. **Transmission**: *The axon terminal releases neurotransmitters into the synaptic cleft, initiating a new chemical signal.*

The Epistemological Bridge

We now know the **reality** side. Let's write it down.

```
[ Biological Reality ]  ======================  Abstraction  ======================>  [ Computational Model ]
 Dendritic signals                               XXX                                    XXX
 Synaptic transmission                           XXX                                    XXX
 Resting / Threshold potential                   XXX                                    XXX
 Soma / Membrane sum                             XXX                                    XXX
 Action Potential (all/none)                     XXX                                    XXX
        XXX                                      XXX                                    XXX
        XXX                                      XXX                                    XXX
        XXX                                      XXX                                    XXX
        XXX                                      XXX                                    XXX
        XXX                                      XXX                                    XXX
        XXX                                      XXX                                    XXX
```

> Five mechanisms on the left, nothing on the right yet. **What do we keep, and how do we write it down?** That is Chapter 3.

Part 1: Foundations • Chapter 3

From Biology to Mathematics

Abstraction: What Do We Keep, What Do We Discard?

Remember: **THE BRIDGE IS A MAP, NOT A MIRROR.** Every abstraction choice is a trade-off.

Kept

- **Dendrite → input:** a channel carrying an incoming value
- **Synaptic strength → weight:** how strongly one input counts
- **Somatic summation → weighted sum:** integration of all inputs
- **Threshold → activation function:** a decision rule on the sum

Discarded

- **Temporal dynamics:** exact spike timing, refractory periods
- **Spatial geometry:** dendrite location, cable properties
- **Neurotransmitter identity:** chemical diversity of synapses
- **Energy constraints:** metabolic cost of firing

Encoding Information: The Firing Rate

A single spike carries almost no information on its own: it's all-or-nothing. So how does the brain encode intensity, e.g., "how loud" or "how bright"?

> **Rate coding:** information is encoded in the FREQUENCY of spikes over a time window, not in the shape or size of any single spike.

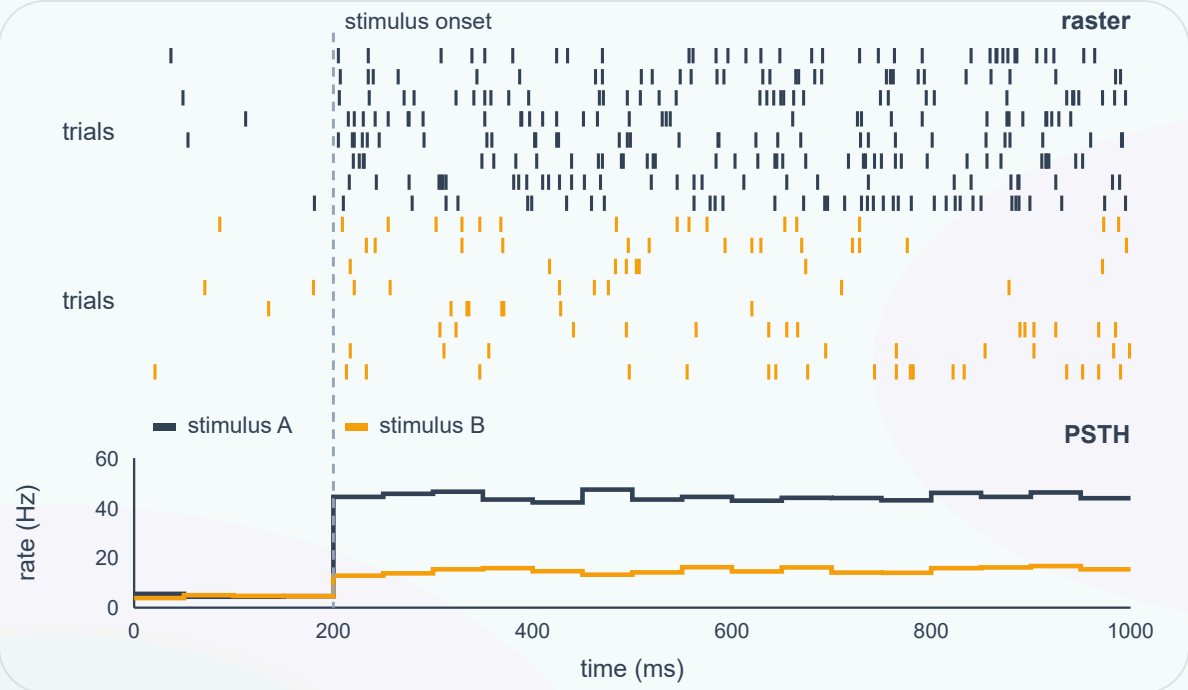
Weak stimulus → few spikes per second

e.g. ~5 Hz

Strong stimulus → many spikes per second

e.g. ~80 Hz

Reading Spike Data: Raster and PSTH



Raster plot: one row per trial, one tick per spike. The same neuron, recorded many times.

PSTH *(peri-stimulus time histogram)*: cut time into bins, count the spikes, average over trials. You get a **firing rate over time**.

> Counting the spikes in a window gives **one number per neuron and per trial**: a firing rate, in Hz.

Rate Is Not the Only Code

Rate coding *(our choice)*

- Information = **how many** spikes in a window
- Simple, robust, easy to compute
- Loses **when** each spike happened
- Needs a **window**: a real-time decoder waits for it to close

Temporal codes *(what we discard)*

- **Latency**: how fast the first spike arrives
- **Timing**: the exact pattern of spikes
- **Synchrony**: neurons firing together

> Choosing rate coding is an **abstraction choice**: two stimuli with the same spike count become **identical** for our model. Keep this in mind for your projects.

TIME TO CODE



notebooks/Chapter3_LiveExercises.ipynb • Part A: Warm-Up

Warm-Up: Simulate a Spike Train

> **</> Check your Python and your maths.** Simulate a neuron, then turn its spikes into a firing rate.

Task: In notebooks/Chapter3_LiveExercises.ipynb, part A:

1. Cut 1 second into **1 ms bins**. In each bin, a neuron firing at r Hz spikes with probability $p = r \times dt$
2. Simulate a **weak** stimulus (5 Hz) and a **strong** one (80 Hz), then **count** the spikes: do you get the rates back?
3. Simulate **3 neurons** at once: an array of shape `(3, 1000)`. Get the vector x of their firing rates
4. Compute $\sum_i w_i x_i$ with $w = (0.5, 0.3, -0.8)$, first with a loop, then with `np.dot`
5. Count the spikes of each neuron in **4 windows of 250 ms**: an array of shape `(3, 4)`

Warm-Up: Simulate a Spike Train

```python
rng = np.random.default_rng(184)
dt = 0.001                          # 1 ms bins
n_bins = 1000                       # 1 second

def spike_train(rate, n_bins, dt, rng):
    p = rate * dt                   # spike probability per bin
    return (rng.random(n_bins) < p).astype(int)

weak = spike_train(5, n_bins, dt, rng)
strong = spike_train(80, n_bins, dt, rng)
print(weak.sum(), strong.sum())  # 4 80: about 5 and 80

rates = np.array([40, 20, 10])   # 3 neurons
spikes = (rng.random((3, n_bins)) < rates[:, None] * dt).astype(int)
x = spikes.sum(axis=1) / (n_bins * dt)   # Hz

w = np.array([0.5, 0.3, -0.8])
print(np.dot(w, x))                 # 17.6: close to 18

windows = np.zeros((3, 4), dtype=int)   # 4 windows of 250 ms
for k in range(4):
    windows[:, k] = spikes[:, k*250:(k+1)*250].sum(axis=1)
```

- A spike train is just an array of **0 and 1**
- `sum` over time gives a **count**; divided by the duration, a **rate**
- `axis=1` sums over time, **one rate per neuron**
- `np.dot(w, x)` **is** $\sum_i w_i x_i$
- Counting in **windows** keeps a coarse time axis: each row still adds up to the total

> You just went from a **simulated biological signal** to the **input of an artificial neuron**.

The Mathematical Neuron: The Weighted Sum

We now have everything to write the soma's spatial summation as an equation.

$$z = \sum_{i=1}^{n} w_i x_i$$

- x_i : input i *(firing rate from presynaptic neuron i)*
- w_i : weight of input i *(synaptic strength)*
- z : pre-activation *(total integrated signal at the axon hillock)*

Live Exercise: The Weighted Sum by Hand

✎ **Paper & pencil: no code.** Use the weighted-sum equation $z = \sum_i w_i x_i$ you just derived.

A postsynaptic neuron receives 3 synapses:

Synapse i	x_i *(firing rate, Hz)*	w_i *(synaptic strength)*
1	40	+0.5
2	20	+0.3
3	10	−0.8

Task 1: Compute z for the table above, by hand.

Task 2: Now imagine the 3 presynaptic neurons all go silent: $x_1 = x_2 = x_3 = 0$. Compute z again.

Task 1: $z = 0.5(40) + 0.3(20) - 0.8(10) = 20 + 6 - 8 = 18$

Task 2: with $x_1 = x_2 = x_3 = 0$, the formula gives $z = 0$, **WHATEVER THE WEIGHTS ARE** . Does that match a real neuron at rest?

Introducing the Bias

With only a weighted sum, **zero input always means zero output.** But a biological neuron never rests at 0 mV, it rests at around **-70 mV**, and must climb to a threshold around **-55 mV** before firing.

Bias (b): a constant term, independent of any input, that shifts the pre-activation, playing the role of the **RESTING POTENTIAL / FIRING THRESHOLD OFFSET** .

$$z = \sum_{i=1}^{n} w_i x_i + b$$

From z to a Decision: What's Still Missing

$$z = \sum_{i=1}^{n} w_i x_i + b$$

z is just a **real number**. It can be -7, 0.3, or 42. On its own, it tells us **nothing about whether the neuron fires.**

Remember the biology: firing is **ALL-OR-NOTHING**. A real neuron either sends a spike, or it doesn't. There is no "half a spike."

The Activation Function

> **Activation function (φ):** a function applied to the pre-activation z that produces the neuron's actual output, $y = \varphi(z)$, playing the role of the **FIRING DECISION**.

- **Input:** z, the pre-activation *(can be any real number)*
- **Output:** y, what the neuron actually sends downstream

There are many possible choices for φ; we'll meet others in Chapter 7. For now, we pick the one that matches biology most directly.

The Step Function: Modeling "All-or-Nothing"

$$\varphi(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

$z \geq 0$: the neuron **FIRES** $\rightarrow y = 1$

$z < 0$: the neuron **STAYS SILENT** $\rightarrow y = 0$

This is the **step function** (or Heaviside function): the mathematical counterpart of the firing threshold, now sitting at $z = 0$ thanks to the bias.

The Complete Artificial Neuron

$$z = \sum_{i=1}^{n} w_i x_i + b \qquad\qquad y = \varphi(z)$$

1. **Inputs** x_i: *firing rates from presynaptic neurons*
2. **Weights** w_i: *synaptic strengths*
3. **Integration:** $z = \sum_i w_i x_i + b$ *(spatial summation + resting offset)*
4. **Decision:** $y = \varphi(z)$ *(the firing threshold, as a step function)*

Live Exercise: From z to y (Paper & Pencil)

> ✎ **Paper & pencil: no code.** Use $z = \sum_i w_i x_i + b$ and the step function φ.

The same postsynaptic neuron as before, now with a bias:

Synapse i	x_i ***(Hz)***	w_i
1	40	+0.5
2	20	+0.3
3	10	−0.8

Bias: $b = -25$

Task 1: Compute z, then $y = \varphi(z)$.
Does the neuron fire?

Task 2 (bonus): What is the smallest value of b that would make it fire instead?

Live Exercise: From z to y (Paper & Pencil)

✎ **Paper & pencil: no code.** Use $z = \sum_i w_i x_i + b$ and the step function φ.

The same postsynaptic neuron as before, now with a bias:

Synapse i	x_i *(Hz)*	w_i
1	40	+0.5
2	20	+0.3
3	10	−0.8

Bias: $b = -25$

Task 1:
$z = 18 - 25 = -7 \;\Rightarrow\; y = \varphi(-7) = 0$
the neuron **stays silent**.

Task 2: firing needs $z \geq 0$, i.e.
$18 + b \geq 0 \;\Rightarrow\; b \geq -18.$